ELECTRODYNAMICS II · SEMESTER 5

Induction and Maxwell equations

Module 2 · PYQ + model paper

40 questions · 20 short + 16 problems + 4 essays
Double the question count of the supplied April 2026 FYUGP prerequisite paper.

CU-FYUGP syllabus 2024 · study edition
SI units · phone pages · October 2026

Practice instructions

Suggested time: two sittings of two hours. Section A: 20 questions × 3 marks, ceiling 48. Section B: 16 × 6, ceiling 72. Section C: answer any two of four × 10. Maximum: 140.

This doubles the supplied April 2026 **Electrodynamics I** 70-mark practice pattern. The current ED II syllabus specifies 70 theory marks but does not supply a question pattern here. This is a study paper, not an official FYUGP mock.

Each numbered question has a distinct primary assessment target. Longer questions combine earlier methods. N25 means November 2025 CBCSS ED I; A26 means April 2026 FYUGP ED I. Tags explicitly identify these older or prerequisite courses. Expanded/adapted means wording, scope or mark allocation was changed. Overlapping past questions are tagged with their original course and number; untagged questions are original model practice.

All four theory modules are included in the set. Module V is laboratory work and is outside this theory-paper series.

Section A · Short answers

3 marks each · ceiling 48

1. State local Ohm's law, define conductivity and distinguish it from resistivity.

Model · syllabus-based

2. Derive the resistance of a uniform wire of length ℓ and area A from local Ohm's law.

Model · syllabus-based

3. Define emf and explain why its name can be misleading.

Model · syllabus-based

4. Can a purely electrostatic field provide a nonzero emf around a closed loop? Explain.

Model · syllabus-based

5. A 0.40 m rod moves at 3.0 m/s perpendicular to a 0.50 T field and to its own length. Find the emf magnitude.

Model · syllabus-based

6. If magnetic forces do no work on individual charges, where does the heat in a moving-rod circuit come from?

Model · syllabus-based

7. State Lenz's rule and distinguish opposition to flux from opposition to flux change.

Model · syllabus-based

8. List three ways to change the flux through a flat loop in uniform B.

Model · syllabus-based

9. Can there be induced E at a location containing no conductor? Give the governing law.

Model · syllabus-based

10. Why cannot a general induced E be described using only a single-valued scalar electrostatic potential?

Model · syllabus-based

11. Define self-inductance and the henry, stating the linearity assumption.

Model · syllabus-based

12. Distinguish the induced emf of an ideal inductor from its passive terminal voltage drop.

Model · syllabus-based

13. Define mutual inductance and explain why coil orientation matters.

Model · syllabus-based

14. Find the energy of a 0.30 H inductor carrying 2.0 A. What happens if current is reversed without changing magnitude?

Model · syllabus-based

15. Write and interpret the local continuity equation.

Model · syllabus-based

16. What is displacement current in vacuum? Does it require charge carriers crossing the surface?

Model · syllabus-based

17. Show that the divergence of a magnetic field vanishes, and state its integral form.

Prerequisite-course PYQ A26 Q15 · adapted marks

18. Write Faraday's law in differential form and explain each side.

Model · syllabus-based

19. State Ampère-Maxwell law in matter and identify the sources on its right side.

Model · syllabus-based

20. Why is normal B continuous at an ordinary interface, even if permeability changes?

Model · syllabus-based

Section B · Problems and explanations

6 marks each · ceiling 72

21. A wire has ℓ=2 m, A=1.0×10^{-6} m² and σ= 5.0×10^{7} S/m. Apply 0.10 V. Find R, I, E, J and total power.

Model · syllabus-based

22. A 12 V source of internal resistance 1 Ω drives a 5 Ω load. Find current, terminal voltage and the power budget.

Model · syllabus-based

23. The rod in Q5 completes a circuit of total resistance 2 Ω. Find current, magnetic drag and pulling power at constant speed.

Model · syllabus-based

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24. A 200-turn coil of area 0.010 m² has its normal along a field that rises uniformly from 0.10 T to 0.40 T in 0.050 s. Find its emf, including sign.

Model · syllabus-based

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25. A coil has N turns and area A and rotates with angular speed ω in uniform B. Derive its sinusoidal emf when its normal initially points along B.

Model · syllabus-based

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26. Uniform $B(t)\hat{z}$ occupies a circular region of radius a. Derive the azimuthal E for r<a, with positive circulation counterclockwise viewed from +z.

Model · syllabus-based

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27. For Q26 find E at r>a and evaluate its magnitude for a=0.10 m, r=0.20 m and $\dot{B} = 4$ T/s.

Model · syllabus-based

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28. Derive L for a long solenoid with N turns, length ℓ, area A and uniform linear permeability μ.

Model · syllabus-based

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29. Two ideal coaxial long coils share area 2×10^{-4} m² and length 0.20 m, with 500 and 1000 turns. In air, find M and the emf magnitude in coil 2 when $dI_1/dt = 3$ A/s.

Model · syllabus-based

30. Derive the stored energy of a constant-L inductor from the work supplied by a source.

Model · syllabus-based

31. Find the vacuum magnetic energy density for B=0.020 T, and the energy in 0.010 m³ of uniform field.

Model · syllabus-based

32. An ideal parallel-plate vacuum capacitor has area A and charging current I. Show that the displacement current through the gap equals I.

Model · syllabus-based

33. Write the four integral Maxwell equations for a fixed loop/surface in vacuum, explaining the source or change in each.

Legacy PYQ N25 Q6 · expanded to Maxwell system

34. Show how changing polarization contributes to current and enters Ampère-Maxwell law in matter.

Model · syllabus-based

35. An interface has normal $\hat{n} = \hat{z}$ from medium 1 to 2 and free surface current $K_f = K\hat{x}$. What tangential H jump is required? Also state the normal B condition.

Model · syllabus-based

36. At a stationary interface with no free surface charge/current, medium 2 has $\epsilon_2 = 2\epsilon_1$, $\mu_2 = 3\mu_1$. Relate all normal and tangential E and H components.

Model · syllabus-based

Section C · Essays

10 marks each · answer any two

37. Define line, surface and volume current measures, derive charge continuity, then explain Maxwell's correction to Ampère's law using a charging capacitor.

Legacy PYQ N25 Q21 · expanded

38. Compare induction in a stationary loop with induction in a moving conductor, including the correct emf expression, sign and energy source.

Model · syllabus-based

39. Derive magnetic energy from circuit work and recover the energy density inside a long linear solenoid. Discuss assumptions.

Model · syllabus-based

40. Derive the four macroscopic electromagnetic boundary conditions at a stationary ordinary interface carrying free surface charge σ_f and free surface current K_f.

Model · syllabus-based

Coverage audit

40 numbered questions; 40 primary targets. The list below makes the intended distinction explicit.

Q1 · **T1** Microscopic Ohm law
Q2 · **T1** Wire resistance scaling
Q3 · **T2** Emf meaning and units
Q4 · **T2** Electrostatic circulation
Q5 · **T3** Rod voltage
Q6 · **T3** Motional energy source
Q7 · **T4** Lenz sign
Q8 · **T4** Changing area and angle
Q9 · **T5** Induced field without wire
Q10 · **T5** Potential limitation
Q11 · **T6** Self-inductance definition
Q12 · **T6** Back emf versus terminal drop
Q13 · **T6** Mutual-inductance direction convention
Q14 · **T7** Inductor energy calculation
Q15 · **T8** Continuity equation
Q16 · **T8** Displacement-current distinction
Q17 · **T9** Gauss magnetic law
Q18 · **T9** Faraday differential law
Q19 · **T10** Free-source Maxwell law
Q20 · **T11** Normal magnetic continuity
Q21 · **T1** Conduction and Joule power
Q22 · **T2** Battery terminal behaviour
Q23 · **T3** Moving rod power balance
Q24 · **T4** Flux ramp in a coil

Q25 · **T4** Rotating flux differentiation

Q26 · **T5** Electric field inside changing-flux disk

Q27 · **T5** Electric field outside changing-flux disk

Q28 · **T6** Solenoid self-inductance

Q29 · **T6** Coupled coils calculation

Q30 · **T7** Work to establish current

Q31 · **T7** Field energy density calculation

Q32 · **T8** Charging capacitor displacement current

Q33 · **T9** Integral Maxwell equations

Q34 · **T10** Polarization current

Q35 · **T11** Boundary jump from a sheet current

Q36 · **T11** Uncharged current-free interface

Q37 · **T8** Derivation of Maxwell correction

Q38 · **T5** Unifying stationary and motional induction

Q39 · **T7** Circuit-to-field magnetic energy

Q40 · **T11** Derivation of all interface conditions

Past-paper accounting

3 distinct questions from the supplied ED I papers are represented at Q17, Q33, Q37. Their course, date and any adaptations are printed beside the questions. Full original wordings and exclusions are in the source inventory.